Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

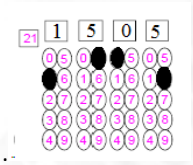
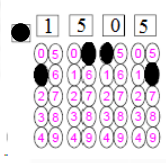
ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN_Shift-1_Batch-II** **Date: 18-01-2026**
Time: 09.00Am to 12.00Pm **GTM-15** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

18-01-2026_Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main-Shift-1_Batch-II_GTM-15_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. **Statement I :** Let T_r be the rth terms of an A.P., for $r = 1,2,3,....$ If for some positive integers m, n we have $T_m = \frac{1}{n}$ and $T_n = \frac{1}{m}$, then T_{mn} is equal to 1

Statement II : For any three positive real numbers a, b and c, $9(25a^2 + b^2) + 25(c^2 - 3ac) = 15b(3a + c)$. Then b, c, a are in A.P.

1) Both Statement I and Statement II are true
2) Both Statement I and Statement II are false
3) Statement I is true but Statement II is false
4) Statement I is false but Statement II is true

2. Number of divisors of $10^3 \times 3^3 \times 7^9$ of the form $4n+1, n \in N$ is

1) 80 **2)** 81 **3)** **79** **4)** 78

3. Let $f(x)$ be a function which satisfy the equation $f(xy) = f(x) + f(y)$ for all $x > 0, y > 0$ such that $f'(1) = 2$. Find the area of the region bounded by the curves $y = f(x), y = \left|x^3 - 6x^2 + 11x - 6\right|$ and $x = 0$

1) $\frac{17}{4}$ **2)** $\frac{11}{4}$ **3)** $\frac{15}{4}$ **4)** $\frac{5}{4}$

4. If $a = \lim\limits_{x \to 0} x^5 \left[\frac{1}{x^3}\right]$ (where $[x]$ is greatest integer function) and

$$b = \lim_{x \to 0} \frac{e^{x^3} - \left(1 - x^3\right)^{\frac{1}{3}} + \left(\left(1 - x^2\right)^{\frac{1}{2}} - 1\right)\sin x}{x \sin^2 x}$$

, then a + 6b is equal to

1) 0 **2) 2** **3) 3** **4) 5**

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

5. $\int \frac{x^2}{(x\cos x - \sin x)(x\sin x + \cos x)}\,dx$ is equal to

1) $\log\left|\frac{x\sin x + \cos x}{x\cos x - \sin x}\right| + C$ **2)** $\log\left|\frac{x\tan x + 1}{x\tan x - 1}\right| + C$

3) $\log\left|\frac{x\cos x + \sin x}{x\sin x - \cos x}\right| + C$ **4)** $\log\left|\frac{x\cot x + 1}{x\cot x - 1}\right| + C$

6. If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is

1) $\sqrt{3}$ **2)** $\frac{3}{\sqrt{2}}$ **3)** $3\sqrt{2}$ **4)** $2\sqrt{3}$

7. Let $f(x) = \sin x + \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} (\sin x + t\cos x) f(t)\,dt$, then match the entries of List I with List II and mark the correct option

List I		List II	
(A)	If range of $f(x)$ is $[\alpha,\beta]$ then $\alpha+\beta$ is	(I)	$\frac{2}{3}$
(B)	$f\left(\frac{3\pi}{2}\right)$ is	(II)	$\frac{1}{\sqrt{2}}$
(C)	$f(\pi)$ is	(III)	$\frac{1}{3}$
(D)	$f\left(\frac{5\pi}{4}\right)$ is	(IV)	0

1) (A) – (III), (B) – (I), (C) – (IV), (D) – II
2) (A) – (III), (B) – (II), (C) – (I), (D) – IV
3) (A) – (I), (B) – (III), (C) – (IV), (D) – II
4) (A) – (IV), (B) – (III), (C) – (I), (D) – II

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

8. Number of eight digit numbers which can be formed using the digits 1, 3, 4, 5, 6, 7, 8, 9 (without repetition) if each number has to be divisible by 275 is

1) 12 **2)** 24 **3)** 36 **4)** 72

9. Let $R=\{(3,3)(5,5),(9,9),(12,12),(5,12),(3,9),(3,12),(3,5)\}$ be a relation on the set $A=\{3,5,9,12\}$. Then, R is :

1) reflexive, symmetric but not transitive

2) symmetric, transitive but not reflexive

3) an equivalence relation

4) reflexive, transitive but not symmetric

10. If the Domain of the function $f(x)=\sin^{-1}\left(\frac{1+x^2}{2x}\right)$ is {a , b} where $a,b\in R$ and $a<b$ and c is the sum of infinite series

$\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+\sin^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{6}}\right)+.....+\sin^{-1}\left(\frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}}\right)+.....\infty$. Then the value of [a + b + c] is [where [.] denotes greatest integer function]

1) 5 **2)** 1 **3)** 3 **4)** 4

11. Let z and ω be two non-zero complex numbers such that $|z|=|\omega|$ and $Arg\,z+Arg\,\omega=\pi$, then z equals

1) ω **2)** $-\omega$ **3)** $\overline{\omega}$ **4)** $-\overline{\omega}$

12. If $(1+ax)^n=1+8x+24x^2+.....+(n+1)^{th}\ terms$ and the lines through $P(a,n)$ touches the circle $x^2+y^2+2x+4y-20=0$ at A and B then length of AB (where $P(a,n)$ is the point of intersection the tangents)

1) $\frac{20}{3}$ **2)** $\frac{40}{3}$ **3)** $\frac{10}{3}$ **4)** $\frac{8}{3}$

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

13. Let $f:R\to(0,\infty)$ be strictly increasing function such that $\lim\limits_{x\to\infty}\dfrac{f(7x)}{f(x)}=1$. Then, the value of $\lim\limits_{x\to\infty}\dfrac{f(5x)}{f(x)}$ is equal to

1) 4 **2)** 1 **3)** $\dfrac{7}{5}$ **4)** 0

14. If $\cos x\dfrac{dy}{dx}-y\sin x=6x,(0<x<\dfrac{\pi}{2})$ and $y\left(\dfrac{\pi}{3}\right)=0$ then $y\left(\dfrac{\pi}{6}\right)$ is equal to

1) $\dfrac{\pi^2}{2\sqrt{3}}$ **2)** $-\dfrac{\pi^2}{2}$ **3)** $-\dfrac{\pi^2}{2\sqrt{3}}$ **4)** $-\dfrac{\pi^2}{4\sqrt{3}}$

15. The polynomial P(z) is the reminder upon dividing z^{2026} by z^2-5z+6. If $P(0)$ can be expressed as $mn(m^r-n^r)$ where m and n are co-prime (m < n) and $r\in N$ then the value of $2n+m$ is λ and x is the solution of the equation $y=2[x]+2$ and $y=3[x-2]$ where $[x]$ denotes the integral part of x then λ is equal to

1) $[x]$ **2)** $\dfrac{1}{[x]}$ **3)** $2[x]$ **4)** $5[x]$

16. Three urns A, B and C contain 4 red, 6 black, 5 red, 5 black and λ red, 4 black balls respectively. One of the urns is selected at random and a ball is drawn. If the ball drawn is red and the probability that it is drawn from urn C is 0.4, then the square of the length of the side of the largest equilateral triangle, inscribed in the parabola $y^2=2\lambda x$ with one vertex at the vertex of the parabola is equal to

1) 12 **2)** 1 **3)** 432 **4)** 1728

17. Let $\vec{a}$ and $\vec{b}$ be the vectors of the same magnitude such that $\dfrac{|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|}{|\vec{a}+\vec{b}|-|\vec{a}-\vec{b}|}=\sqrt{3}+2$.

Then $\dfrac{|\vec{a}+\vec{b}|^2}{|\vec{a}|^2}$ is

1) 2 **2)** $2+\sqrt{2}$ **3)** 3 **4)** $4+2\sqrt{2}$

Sri Chaitanya Educational Institutions
Infinity Learn
40


For More Join: @IITJEE_Advanced

18. If $\{p\}$ denotes the fractional part of the number p, then $\left\{\frac{3^{200}}{8}\right\}$ is equal to

1) $\frac{5}{8}$ **2)** $\frac{7}{8}$ **3)** $\frac{3}{8}$ **4)** $\frac{1}{8}$

19. Let the co-ordinates of one vertex of ΔABC be $A(0,2,\alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha\in\mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^4 is equal to

1) 3 **2)** 27 **3)** 81 **4)** 243

20. If $A=\begin{bmatrix}3 & 1\\ 4 & 2\end{bmatrix}$ and B_p is another matrix defined as $B_p = A^p\underbrace{\left(adj\left(adj.....(adj.A)\right)\right)}_{p-times}$ where $p\in N$, then identify the wrong statement

1) $tr.(B_1)=4$ **2)** $\det.(B_2)=8$

3) $\sum_{p=1}^{\infty}\det.(B_p^{-1})=1$ **4)** $adj(adjB_2)=A^3$

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. If $f(x)$ is an odd function on R, $f(1)=1$ and $f\left(\frac{x}{x-1}\right)=xf(x)$ for any $x<0$, then the value of $f(1)f\left(\frac{1}{100}\right)+f\left(\frac{1}{2}\right)f\left(\frac{1}{99}\right)+f\left(\frac{1}{3}\right)f\left(\frac{1}{98}\right)+.....+f\left(\frac{1}{50}\right)f\left(\frac{1}{51}\right)$ is $\frac{2^K}{(K+1)!}$ then K is equal to

Sri Chaitanya Educational Institutions Infinity Learn 40 YEARS



For More Join: @IITJEE_Advanced

22. Let $f(x)=2^x-x^2, x\in R$ If m and n are respectively the number of points at which the curves $y=f(x)$ and $y=f'(x)$ intersects the $x-$ axis, then the value of $(m+n)^2$ is equal to

23. An online exam is attempted by 50 candidates out of which 20 are boys. The average marks obtained by boys is 12 with a variance 2. The variance of marks obtained by 30 girls is also 2. The average marks of all 50 candidates is 15. If μ is the average marks of girls and σ^2 is the variance of marks of 50 candidates then $\mu^2+\sigma^2$ is equal to

24. If $\int_0^{\frac{\pi}{4}} \frac{\sin x+\cos x}{9+16\sin 2x}dx=\frac{1}{k}\log_e a$ then $k+a$ is equal to

25. Let $R=\left\{\begin{pmatrix} a & 3 & b \\ c & 2 & d \\ 0 & 5 & 0 \end{pmatrix}; a,b,c,d\in\{0,3,5,7,11,13,17,19\}\right\}$. Then the number of invertible matrices in R is

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. A Physical quantity Q is found to depend on quantities a, b, c by the relation $Q = \frac{a^4 b^3}{c^2}$.

The percentage error in a, b and c are 1 %, 2 % and 5 % respectively. Then, the percentage error in Q is :

1) 60 % **2)** 20 % **3)** 30 % **4)** 40 %

27. Statement I : In YDSE, if intensity of each source I_o (coherent Sources), then minimum and maximum intensity is zero and $4I_o$ respectively

Statement II : In YDSE energy conservation is not followed

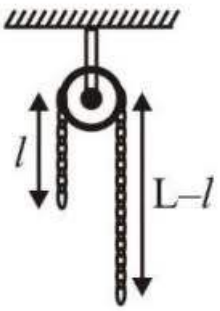
1) Statement I is true and Statement II is false

2) Both Statement I & II are false

3) Both Statement I & II are true

4) Statement I is false and Statement II is true

28. A uniform metal chain of mass 'M' and length 'L' passes over a massless and frictionless pulley. It is released from rest with a part of its length 'l' is hanging on one side and rest of its length 'L – l' is hanging on the other side of the pulley. At a certain point of time, when $l = \frac{L}{x}$, the acceleration of the chain is $\frac{g}{3}$. The value of x is _____.



1) 6 **2)** 2 **3)** 4 **4)** 3

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

29. Moment of inertia of rod of length l and mass m about one end of a rod is

1) $\frac{ml^2}{12}$ **2)** $\frac{ml^2}{3}$ **3)** $\frac{ml^2}{2}$ **4)** $\frac{3}{2}ml^2$

30. A satellite is revolving the round the earth in an orbit of radius r with time period T. If the satellite is revolving round the earth in an orbit of radius $r+\Delta r(\Delta r \ll r)$ with time period $T+\Delta T(\Delta T \ll T)$ then,

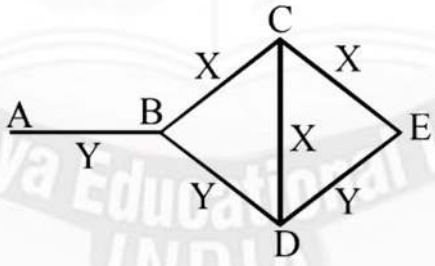
1) $\frac{\Delta T}{T}=\frac{3}{2}\frac{\Delta r}{r}$ **2)** $\frac{\Delta T}{T}=\frac{2}{3}\frac{\Delta r}{r}$ **3)** $\frac{\Delta T}{T}=\frac{\Delta r}{r}$ **4)** $\frac{\Delta T}{T}=3\frac{\Delta r}{r}$

31. A uniform cylindrical rod of length L and radius *r*, is made from a material, whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of linear expansion of the material of the rod, is (nearly) equal to

1) $9F/\left(\pi r^2 YT\right)$ **2)** $6F/\left(\pi r^2 YT\right)$ **3)** $F/\left(\pi r^2 YT\right)$ **4)** $F/\left(3\pi r^2 YT\right)$

32. Three rods of material X and three rods of material Y are connected as shown in the figure. All the rods are of identical length and cross sectional area. If the end A is maintained at $60^\circ C$ and the junction E at $10^\circ C$, find the effective thermal resistance.

Given : Length of the rod = L , Area of cross section = A, thermal conductivity of X = K, thermal conductivity of Y = 2K



1) $\frac{4L}{3KA}$ **2)** $\frac{7L}{6KA}$ **3)** $\frac{4KA}{3L}$ **4)** $\frac{7KA}{3L}$

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

33. The relation between root mean square speed (V_{rms}) and most probable speed (V_p) for the molar mass M of oxygen gas molecules at the temperature of 300 K will be

1) $v_{rms} = \sqrt{\frac{2}{3}} v_p$ **2)** $v_{rms} = \sqrt{\frac{3}{2}} v_p$ **3)** $v_{rms} = v_p$ **4)** $v_{rms} = \sqrt{\frac{1}{3}} v_p$

34. A travelling wave is described by the equation

$$y(x,t) = [5\sin(8x - 4t)]m$$

The velocity of the wave is : [all the quantities are in SI unit]

1) $4ms^{-1}$ **2)** $2ms^{-1}$ **3)** $0.5ms^{-1}$ **4)** $8ms^{-1}$

35. Assertion (A) : NAND or NOR gates are called digital building blocks.
Reason (R) : The repeated use of NAND or NOR gates can produce all the basic and complex gates

1) Both (A) and (R) are correct ands (R) is the correct explanation of (A)

2) Both (A) and (R) are correct ands (R) is not the correct explanation of (A)

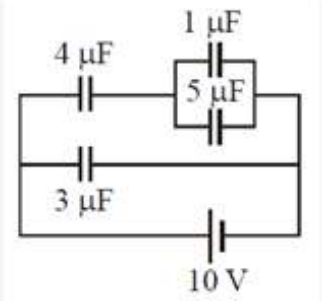
3) (A) is not correct but (R) is correct

4) (A) is correct but (R) is not correct

36. A certain charge Q is divided into two parts q and (Q – q) . How should the charges Q and q be divided so that q and (Q – q) placed at a certain distance apart experience maximum electrostatic force?

1) $Q = \frac{q}{2}$ **2)** $Q = 2q$ **3)** $Q = 4q$ **4)** $Q = 3q$

37. In the given circuit, the charge on 4 μF capacitor will be :

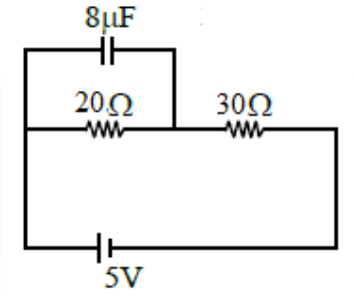


1) 5.4 μC **2)** 9.6 μC **3)** 13.4 μC **4)** 24 μC

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

38. At steady state the charge on the capacitor, as shown in the circuit below is _____ μC.



1) 16 **2)** 8 **3)** 24 **4)** 32

39. A deuteron and an alpha particle having equal kinetic energy enter perpendicular into a magnetic field. Let r_d and r_α be their respective radii of circular path. The value of is $\frac{r_d}{r_\alpha}$ equal to

1) $\frac{1}{\sqrt{2}}$ **2)** $\sqrt{2}$ **3)** **1** **4)** **2**

40. Match List I with List II

	List I		List II
(a)	UV rays	(i)	Diagnostic tool in medicine
(b)	X – rays	(ii)	Water purification
(c)	Microwave	(iii)	Communication, Radar
(d)	Infrared wave	(iv)	Improving visibility in foggy days

Choose the correct answer from the options given below:

1) (a) – (iii), (b) – (ii), (c) – (i), d – (iv)

2) (a) – (ii), (b) – (i), (c) – (iii), d – (iv)

3) (a) – (ii), (b) – (iv), (c) – (iii), d – (i)

4) (a) – (iii), (b) – (i), (c) – (ii), d – (iv)

41. Force acting per unit length between two current carrying conductors i_1, i_2 are separated by a distance d is

1) $\frac{F}{l}=\frac{\mu_0 i_1 i_2}{2\pi d}$ **2)** $\frac{F}{l}=\frac{\mu_0 i_1 i_2}{4\pi d}$ **3)** $\frac{F}{l}=\frac{\mu_0 i_1 i_2}{\pi d}$ **4)** $\frac{F}{l}=\frac{\mu_0 i_1 i_2}{5\pi d}$

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

42. The relationship between the magnetic susceptibility (χ) and the magnetic permeability (μ) is given by:

$(\mu_o$ **is the permeability of free space and** μ_r **is relative permeability**$)$

1) $\mu_r = 1+\chi$ **2)** $\chi = \frac{\mu_r}{\mu_o}+1$ **3)** $\chi = \mu_r + 1$ **4)** $\chi = 1-\frac{\mu_r}{\mu_0}$

43. Assertion (A) : When a body dropped from a height exploded in midair is, its centre of mass keeps moving in vertically downward direction.

Reason (R) : Explosion occurs under internal forces only. External forces zero.

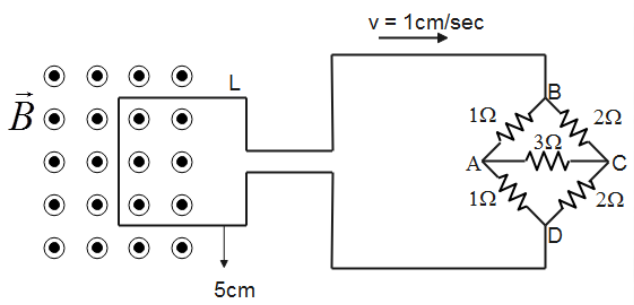
1) Both (A) and (R) are correct ands (R) is the correct explanation of (A)

2) Both (A) and (R) are correct ands (R) is not the correct explanation of (A)

3) (A) is not correct but (R) is correct

4) (A) is correct but (R) is not correct

44. The figure shows a square loop L of side 5 cm which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of 1 cm s^{-1}. At some instant, a part of L is in a uniform magnetic field of 1 T, perpendicular to the plane of the loop. If the resistance of L is 2.7Ω. The current in the loop at that instant will be close to



1) $60\mu A$ **2)** $170\mu A$ **3)** $150\mu A$ **4)** $125\mu A$

45. Statement I : Electrical potential may exist at a point where the electrical field is zero

Statement II : Electrical field may exist at a point where the electrical potential is zero

1) Statement I and Statement II are true

2) Both Statement I and II are false

3) Statement I true and Statement II false

4) Statement I false and Statement II true

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
For More Join: @IITJEE_Advanced

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A particle executes SHM of amplitude A. The distance from the mean position when it's kinetic energy becomes equal to its potential energy is $\frac{A}{\sqrt{x}}$, then find x

47. An AC circuit has an inductor and a resistor of resistance R in series, such that $X_L = 3R$ Now, a capacitor is added in series such that $X_C = 2R$. The ratio of new power factor with the old power factor of the circuit is $\sqrt{5} : x$. The value of x is _____

48. A point object is placed at a distance of 12cm on the axis of a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed at a distance of 10cm from the lens, such that the image formed by the combination coincides with the object itself. What the focal length of the convex mirror?

49. A Young's double-slit experiment is performed using monochromatic light of wavelength λ. The intensity of light at a point on the screen, where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{6}$ is given by $\frac{nK}{4}$, where n in an integer. The value of n is

50. In the hydrogen spectrum, λ the wavelength of first transition line of Lyman series. The wavelength difference will be "aλ" between the wavelength of 3rd transition line of Paschen series and that of 2nd transition line of Balmer series where a = _____

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Match List I with the List II

List - I (Reaction)		List - II (Type of redox reaction)	
A	$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$	I	Decomposition
B	$2Pb(NO_3)_{2(s)} \rightarrow 2PbO_{(s)} + 4NO_{2(g)} + O_{2(g)}$	II	Displacement
C	$2Na_{(s)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + H_{2(g)}$	III	Disproportionation
D	$2NO_{2(g)} + 2OH^-_{(aq)} \rightarrow NO^-_{2(aq)} + NO^-_{3(aq)} + H_2O_{(l)}$	IV	Combination

Choose the correct answer from the options given below

1) (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

2) (A) – (III), (B) – (II), (C) – (I), (D) – (IV)

3) (A) – (II), (B) – (III), (C) – (IV), (D) – (I)

4) (A) – (IV), (B) – (I), (C) – (II), (D) – (III)

52. Statement I : The pentavalent oxide of group 15 element, E_2O_5 is more acidic than trivalent oxide, E_2O_3 of the same elements.

Statement II : The acidic character of trivalent oxide of group 15 elements E_2O_3 **decreases** down the group

1) Both Statement I and Statement II are true

2) Both Statement I and Statement II are false

3) Statement I is true but Statement II is false

4) Statement I is false but Statement II is true

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

53. Assertion (A) : Cu^{2+} in water is more stable than Cu^{+}

Reason (R) : Enthalpy of hydration for Cu^{2+} is much less than that of Cu^{+}

1) Both (A) and (R) are correct ands (R) is the correct explanation of (A)

2) Both (A) and (R) are correct ands (R) is not the correct explanation of (A)

3) (A) is not correct but (R) is correct

4) (A) is correct but (R) is not correct

54. The coordination environment of Mg^{+2} ion in its complex with $EDTA^{4-}$ is :

1) tetrahedral **2)** octahedral **3)** square planar **4)** Trigonal prismatic

55. The isoelectronic set of ions is

1) Na^{3-}, Li^{+}, Mg^{2+} **and** O^{2-} **2)** Li^{+}, Na^{+}, O^{2-} **and** F^{-}

3) F^{-}, Li^{+}, Na^{+} **and** Mg^{2+} **4)** P^{-3}, S^{-2}, Cl^{-1} **and** Ca^{+2}

56. Which of the following pair of molecule contain odd electron molecule and an expanded octet molecule?

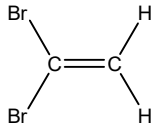
1) BCl_3 **and** SF_6 **2)** NO_2 **and** H_3PO_4 **3)** SF_6 **and** H_2SO_4 **4)** BCl_3 **and** NO

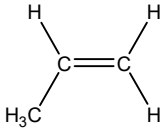
57. Compound that will give positive Lassaigne's test for both nitrogen and sulphur is

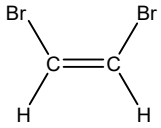
1) $N_2H_4.HCl$ **2)** $NH_2-\overset{\overset{S}{\|}}{C}-NH-NH_2$

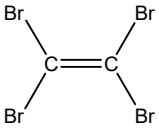
3) NH_4Cl **4)** $NH_2OH.HCl$

58. Which one of the following will show geometrical isomerism?

1) 

2) 

3) 

4) 

59. The correct stability order of carbocations is

1) $(CH_3)_3C^{+} > CH_3-\overset{+}{C}H_2 > (CH_3)_2\overset{+}{C}H > \overset{+}{C}H_3$

2) $\overset{+}{C}H_3 > (CH_3)_2\overset{+}{C}H > CH_3-\overset{+}{C}H_2 > (CH_3)_3\overset{+}{C}$

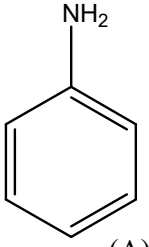
3) $(C_6H_5)_3\overset{+}{C} > (C_6H_5)_2\overset{+}{C}H > C_6H_5-\overset{+}{C}H_2$

4) $\overset{+}{C}H_3 > CH_3-\overset{+}{C}H_2 > (CH_3)_2-\underset{CH3}{\overset{+}{C}}H > (CH_3)_3\overset{+}{C}$

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

60. The increasing order of basicity of the following compounds is

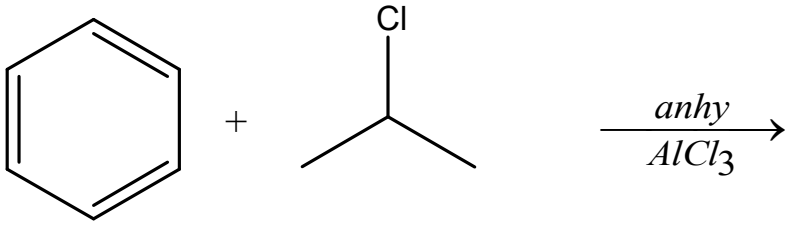


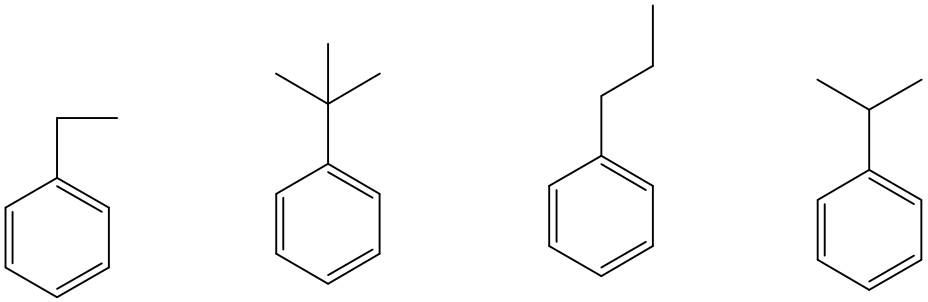
(A) , NH_3 (B) , $R-NH_2$ (C)

1) (A) = (B) = (C) **2)** (A) > (B) > (C)

3) (A) < (B) < (C) **4)** (B) < (A) < (C)

61. The major product 'P' formed in the given reaction is :





1) **2)** **3)** **4)**

62. Consider the given chemical reaction :

$$CH_3-CH=CH-CH_3 \xrightarrow[Heat]{KMnO_4-H_2SO_4} \text{Product "A"}$$

Product "A" is

1) picric acid **2)** oxalic acid **3)** acetic acid **4)** succinic acid

63. In the given reactions identify A and B

$$CH_3-C\equiv C-CH_3+H_2 \xrightarrow{Pd/C} A$$

$$CH_3-C\equiv C-CH_3+H_2 \xrightarrow{Na/Liquid\ NH_3} "B"$$

1) **A :** cis - 2 – Butene B : trans – 2 – butene

2) **A :** trans - 2 – Butene B : cis – 2 – butene

3) A : trans - 2 – Butene B : trans – 2 – butene

4) A : cis – 2 - Butene B : cis – 2 – butene

Sri Chaitanya Educational Institutions Infinity Learn 40 YEARS

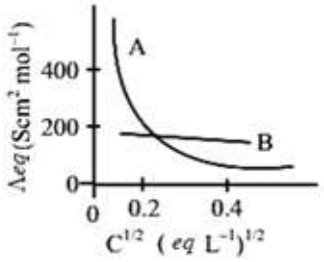


For More Join: @IITJEE_Advanced

64. If a substance of 'A' dissolves in solution of a mixture of 'B' and 'C' with their respective number of moles as n_A, n_B **and** n_C mole fraction of B in the solution is

1) $\frac{n_C}{n_A \times n_B \times n_C}$ **2)** $\frac{n_C}{n_A + n_B + n_C}$ **3)** $\frac{n_C}{n_A - n_B - n_C}$ **4)** $\frac{n_B}{n_A + n_B + n_C}$

65. The equivalent conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are

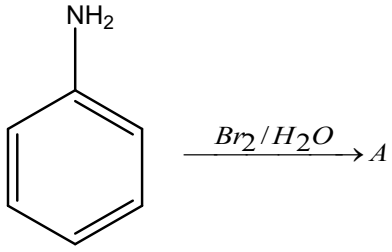


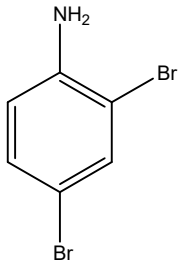
	A	B
1)	Weak electrolyte	Weak electrolyte
2)	Strong electrolyte	Strong electrolyte
3)	Weak electrolyte	Strong electrolyte
4)	Strong electrolyte	Weak electrolyte

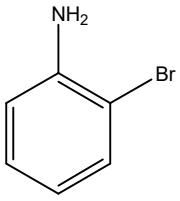
66. Which of the following amino acid containing secondary amine group?

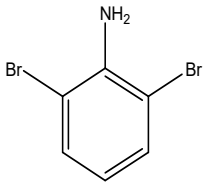
1) Proline **2)** Glycine **3)** Tyrosine **4)** Alanine

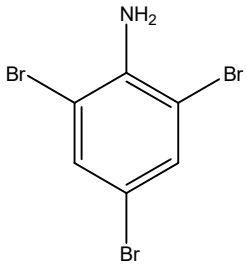
67. Consider the below reaction, the product 'A' is



1) 

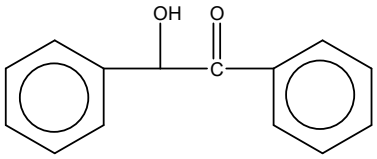
2) 

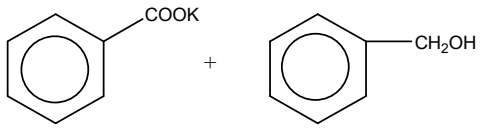
3) 

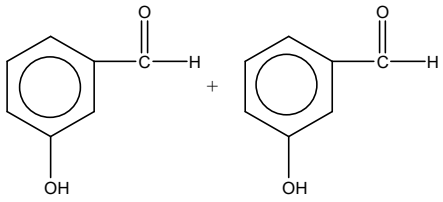
4) 

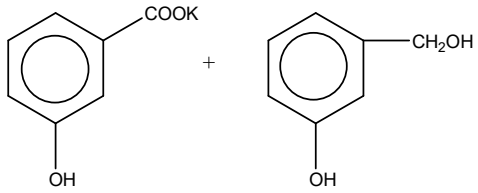
Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS
THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHAIVALAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
For More Join: @IITJEE_Advanced

68. Benzaldehyde on treatment with 50 % KOH solution yields

1) 

2) 

3) 

4) 

69. Statement I : The electronegativity of group 14 elements from Si to Pb gradually increases

Statement II : Group 14 contains non – metallic, metallic, as well as metalloid elements

1) Statement I is false but Statement II is true

2) Statement I is true but Statement II is false

3) Both Statement I and Statement II are true

4) Both Statement I and Statement II are false

70. Assertion(A):Both rhombic and monoclinic Sulphur exist as S_8 while oxygen exists as O_2

Reason (R): Oxygen forms stable $p\pi - p\pi$ multiple bonds with itself and other elements having small size and high electronegativity like C, N, which is not possible for Sulphur

1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

3) (A) is correct but (R) is not correct

4) (A) is not correct but (R) is correct

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

71. At 298 K

$N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g), K_1=4\times10^5$

$N_2(g)+O_2(g)\rightleftharpoons 2NO(g), K_2=1.6\times10^{12}$

$H_2(g)+\frac{1}{2}O_2(g)\rightleftharpoons H_2O(g), K_3=1.0\times10^{-13}$

Based on above equilibria, the equilibrium constant of the reaction

$2NH_3(g)+\frac{5}{2}O_2(g)\rightleftharpoons 2NO(g)+3H_2O(g)$ **is** _____ $\times10^{-33}$ **(Nearest integer)**

72. pH value of 0.001 M KOH solution is _____

73. The minimum number of moles of O_2 required for complete combustion of 1 mole of propane and 2 moles of ethane is _____

74. Assume that the radius of the first Bohr orbit of hydrogen atom is 0.6 $\mathring{A}$. The radius of the third Bohr orbit Li^{+2} of is ------- picometer. (Nearest Integer)

75. A system absorbs 20 kJ of heat and also does 15 kJ work. The ΔU in kJ is

Sri Chaitanya Educational Institutions
Infinity Learn
40 YEARS


For More Join: @IITJEE_Advanced

Sri Chaitanya Educational Institutions | Infinity Learn

Sri Chaitanya Techno School
The right mentor for IIT (JEE), NEET, Olympiad & all Other Competitive exams

40 Years of Excellence 1986-2025

JEE MAIN 2025

31 STUDENTS BELOW 100 AIR









BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS









BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

For More Join: @IITJEE_Advanced